privileged inputs: decide what to generate and how, never enter a trajectory

site map — controls c + their requests

dry runs — data changes $\rightarrow$ κ

human demos — spans H_m of m

Summarize — LLM: $H_m \rightarrow$ guide G_m

Sample — (w, c, x, s_0) — site, control, argument, start

Suggest — $g,\ \kappa$ — instruction + expected check

Execute — $\tau = (o_t, a_t)_{t \le T}$ — LLM script with G_m: DOM $\rightarrow$ mouse, keys

Gate — $\kappa(s_0) = 0,\ \kappa(s_T) = 1$ — keep in $\mathcal{D}_{\text{gen}}$

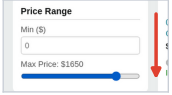
G_m

"Adjust the Max Price: Any slider to 1650."

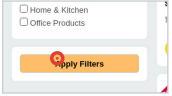
`check: GET ?max_price ∈ [1640, 1670]`



1 `click (184, 444)`



2 `scroll dy=600`



3 `click (89, 284)`

✓ `max_price =1650`

executor script (excerpt)

```
s = dom.one(css=task["target_css"])
act.click(s, at=(frac, 0.5))
...
act.click(btn)
expect.backend()
```

`dom`: read-only queries
`act`: mouse and keys at jittered points on visible elements
`expect`: the backend check

3,128 kept trajectories · 52 training sites · ≈$0.04 each

A trajectory holds only screenshots and pixel actions; failed attempts are kept as negatives.

Planner π_0 (untrained base model)

sees o (URL, elements, screenshot), cannot act; each turn emits $u_t = (\gamma, \iota)$ to delegate, or done(α)

$\mathcal{D}$: 27.6K steps

replayed + augmented

pooled adapter Δ_{pool}

every action family: navigation, text, selection, forms, I/O, drag

executor $\pi_0 + \Delta_{\text{pool}}$, trained on all of $\mathcal{D}$; reasoning steps run on π_0

Held-out SnapLink task, solved in 20 steps:

"Search your shortened links for "archived" and delete the matching links. Then create a short link for https://facebook.com."

1	nav	Click 'My Links'	3 steps
2	text	Type 'archived' in the search field	4 steps
3	nav	Open 'Old Blog Post (Archived)'	3 steps
4	select	Click the red 'Delete' button	3 steps
5	nav	Click 'Create'	3 steps
6	text	Type 'https://facebook.com' into the URL field	4 steps
	done	✓ verifier: links deleted, new link created	